

Higher-Order Cancellation in Exponential Approximants: An Explicit $O(n^{-4})$ Telescoping Family

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September 2026 (V5)

Abstract

We study elementary sequences (x_n) converging to exponential constants e^c for which the successive differences $D_n := x_{n+1} - x_n$ form a rapidly decaying telescoping tail. For any convergent sequence $(x_n) \rightarrow L$, the identity $L = x_1 + \sum_{n \geq 1} D_n$ reduces quantitative convergence questions to asymptotics of D_n ; in particular, if $D_n = Kn^{-(k+1)} + O(n^{-(k+2)})$ then $L - x_N = \frac{K}{k}N^{-k} + O(N^{-(k+1)})$. Within the constrained ansatz

$$x_n = \left(1 + \frac{a_1}{n} + \frac{a_2}{n^2}\right)^{cn+\gamma},$$

we use logarithmic expansions and coefficient cancellation to enforce higher-order decay of D_n . The methodological ingredients are classical; the point here is that, for this specific quadratic-base/affine-exponent competitor class, the order-3 constraints admit an explicit closed-form solution.

Our main result is a two-branch family achieving telescoping order 3 (i.e. $|D_n| = \Theta(n^{-4})$) for any real $c \neq 0$:

$$x_n^{(\text{I})} = \left(1 + \frac{1}{n} + \frac{\sqrt{3}}{6n^2}\right)^{cn+c(\frac{1}{2}-\frac{\sqrt{3}}{6})}, \quad x_n^{(\text{II})} = \left(1 + \frac{1}{n} - \frac{\sqrt{3}}{6n^2}\right)^{cn+c(\frac{1}{2}+\frac{\sqrt{3}}{6})}.$$

We prove $\lim_{n \rightarrow \infty} x_n^{(\cdot)} = e^c$ and derive explicit signed asymptotics

$$D_n^{(\cdot)} \sim -\frac{3e^c A_3^{(\cdot)}}{n^4}, \quad A_3^{(\text{I})} = \frac{c(2\sqrt{3}-3)}{72}, \quad A_3^{(\text{II})} = -\frac{c(3+2\sqrt{3})}{72},$$

implying the tail error $|x_N - e^c| = \Theta(N^{-3})$ for $c \neq 0$. (At $c = 0$ both sequences are identically 1 and the asymptotic statements are vacuous.)

1 Introduction

1.1 Telescoping representations

For any convergent sequence $(x_n) \rightarrow L$, we have the telescoping identity

$$L = x_1 + \sum_{n=1}^{\infty} (x_{n+1} - x_n) = x_1 + \sum_{n=1}^{\infty} D_n.$$

The speed of convergence is governed by the decay of the differences D_n ; faster decay of D_n yields a smaller tail

$$R_N := \sum_{n > N} D_n = L - x_{N+1}.$$

1.2 Positivity of the bases (well-defined real powers)

Because we use real (not complex) exponentiation, we record a positivity check ensuring the bases remain positive for all $n \geq 1$.

Lemma 1.1 (Positivity of the bases). *For all integers $n \geq 1$,*

$$1 + \frac{1}{n} + \frac{\sqrt{3}}{6n^2} > 0 \quad \text{and} \quad 1 + \frac{1}{n} - \frac{\sqrt{3}}{6n^2} > 0.$$

Hence $x_n^{(\text{I})}$ and $x_n^{(\text{II})}$ are well-defined as real numbers for all $n \geq 1$.

Proof. The first inequality is immediate since $1 + \frac{1}{n} + \frac{\sqrt{3}}{6n^2} > 1$. For the second, for $n \geq 1$ we have $n^{-2} \leq 1$, hence $1 + \frac{1}{n} - \frac{\sqrt{3}}{6n^2} \geq 1 - \frac{\sqrt{3}}{6} = \frac{6-\sqrt{3}}{6} > 0$. $\square$

1.3 From difference decay to error decay

The following standard comparison makes the meaning of “telescoping order” explicit.

Lemma 1.2 (Tail rate from difference rate). *Let (x_n) converge to L and set $D_n := x_{n+1} - x_n$. Suppose that for some $k \geq 1$,*

$$D_n = Kn^{-(k+1)} + O(n^{-(k+2)}) \quad (K \neq 0).$$

Then

$$L - x_N = \sum_{n \geq N} D_n = \frac{K}{k} N^{-k} + O(N^{-(k+1)}).$$

In particular, telescoping order k (i.e. $D_n = \Theta(n^{-(k+1)})$) corresponds to an $O(N^{-k})$ convergence rate for the error $|x_N - L|$.

Proof. This follows from the telescoping identity and the asymptotic evaluation of the tail of a p -series, $\sum_{n \geq N} n^{-(k+1)} = \frac{1}{k} N^{-k} + O(N^{-(k+1)})$. $\square$

Remark 1.3 (Signed asymptotics vs. absolute bounds). Lemma 1.2 is stated with a *signed* asymptotic for D_n ; the conclusion $L - x_N = \Theta(N^{-k})$ uses that sign. A bare absolute bound $|D_n| = \Theta(n^{-(k+1)})$ does *not* by itself force a Θ tail rate: e.g. $x_n = 1 + (-1)^n n^{-4}$ has $|D_n| = \Theta(n^{-4})$ but $|x_n - 1| = n^{-4}$, not $\Theta(n^{-3})$. Throughout this paper the relevant families admit genuine signed power-series expansions, so Lemma 1.2 applies directly.

1.4 Why telescoping matters (context and applications)

Although the present paper focuses on elementary closed forms for e^c , the telescoping viewpoint is broadly useful as a way to translate convergence into an explicit remainder model. Many classic acceleration techniques can be viewed as constructing and cancelling leading terms in an error expansion, and the NIST DLMF provides standard language and examples for sequence transformations that accelerate convergence (Aitken, Shanks/Wynn- ϵ , Levin–Weniger, etc.) [6] (see also [7]). In a more recent probabilistic framing, Oates–Karvonen et al. propose probabilistic Richardson extrapolation, where an error model is fitted/propagated to accelerate convergence while quantifying uncertainty [8, 9]. In this light, explicit telescoping families with known D_n asymptotics provide a clean laboratory for validating order-detection by regression and illustrating how algebraic constraints enforce higher-order cancellation in a controlled setting.

1.5 Classical examples and the classical design principle

The prototypical example is Euler’s sequence $(1 + 1/n)^n \rightarrow e$. More generally, for $(1 + a/n)^{bn} \rightarrow e^{ab}$, a direct expansion of the logarithm (Proposition 3.1 below) gives

$$D_n = \frac{e^{ab} a^2 b}{2n^2} + O(n^{-3}), \quad (1)$$

so the leading term vanishes *iff* $a = 0$ or $b = 0$ (both of which give the constant sequence 1). In particular $(1 + c/n)^{cn}$ has limit e^{c^2} and difference order exactly $\Theta(n^{-2})$ for $c \neq 0$; at $c = 1$ it is Euler’s sequence. (An earlier draft of this work asserted the stronger and incorrect claim that the leading term cancels at $a = b$; see Remark 3.2.) The methodology used throughout this paper—taking logarithms, expanding in powers of $1/n$, and enforcing coefficient cancellations to increase decay order—is standard asymptotic-expansion practice; see, for example, [5].

1.6 Competitor class and novelty boundary

Definition 1.4 (Competitor class: quadratic base, affine exponent). We call a sequence *quadratic-base, affine-exponent* if it has the form

$$x_n = \left(1 + \frac{a_1}{n} + \frac{a_2}{n^2}\right)^{cn+\gamma},$$

with parameters a_1, a_2, γ given in elementary closed form (and c a prescribed real constant).

Within this competitor class, telescoping-order constraints reduce to a finite algebraic system in (a_1, a_2, γ) (after the normalizations described later). The novelty boundary in this paper is therefore not the existence of coefficient matching, but the fact that in the quadratic-base, affine-exponent class one can solve the order-3 constraints explicitly in closed form and obtain a two-branch parameterization together with explicit leading asymptotic constants.

1.7 Relation to prior work and novelty (conservative framing)

Refined investigations of exponential constants and limiting processes (including e and the Euler–Mascheroni constant γ) have a long history; see Glaisher [2], Adams [3], and Mitchell–Strain [4]. At a methodological level, logarithmic expansions and cancellation arguments are classical; see Brent [1] and [5]. Modern convergence-acceleration context can be found in the DLMF overview [6] and Corless [7]. The present work also builds on the telescoping-difference framework developed in [12].

Accordingly, the novelty of this paper does not lie in asymptotic expansions per se, but in producing a fully explicit closed-form order-3 telescoping family inside the tightly constrained competitor class of Definition 1.4, including explicit signed leading coefficients controlling D_n and a direct implication for the N^{-3} error rate.

1.8 Main result

Theorem 1.5 (Explicit order-3 family in a quadratic-base, affine-exponent ansatz). *Fix $c \in \mathbb{R} \setminus \{0\}$. Define*

$$x_n^{(\text{I})} := \left(1 + \frac{1}{n} + \frac{\sqrt{3}}{6n^2}\right)^{cn+c(\frac{1}{2}-\frac{\sqrt{3}}{6})}, \quad x_n^{(\text{II})} := \left(1 + \frac{1}{n} - \frac{\sqrt{3}}{6n^2}\right)^{cn+c(\frac{1}{2}+\frac{\sqrt{3}}{6})}.$$

Then for each family:

1. $\lim_{n \rightarrow \infty} x_n^{(\cdot)} = e^c$.
2. $D_n^{(\cdot)} := x_{n+1}^{(\cdot)} - x_n^{(\cdot)}$ satisfies $|D_n^{(\cdot)}| = \Theta(n^{-4})$.

More precisely,

$$D_n^{(\cdot)} \sim -\frac{3e^c A_3^{(\cdot)}}{n^4} \quad \text{as } n \rightarrow \infty, \quad (2)$$

where

$$A_3^{(\text{I})} = \frac{c(2\sqrt{3}-3)}{72}, \quad A_3^{(\text{II})} = -\frac{c(3+2\sqrt{3})}{72},$$

so $K^{(\cdot)} = 3e^c |A_3^{(\cdot)}|$.

Remark 1.6 (Branch labeling and sign). We label branches by the magnitude of the leading asymptotic coefficient: $x_n^{(\text{I})}$ has smaller $|D_n|$ (faster convergence) and $x_n^{(\text{II})}$ has larger $|D_n|$ (slower convergence), since $0 < 2\sqrt{3}-3 < 3+2\sqrt{3}$. For $c > 0$ the two leading errors have *opposite signs*: $x_n^{(\text{I})}$ approaches e^c from above (decreasing), while $x_n^{(\text{II})}$ approaches e^c from below (increasing); the signs reverse for $c < 0$.

2 Framework and Asymptotic Expansion

2.1 General setup

We consider sequences of the form

$$x_n = f(1/n)^{g(n)}, \quad f(t) = 1 + \alpha_1 t + \alpha_2 t^2 + \cdots, \quad g(n) = \beta n + \gamma.$$

Definition 2.1 (Telescoping order). A sequence $(x_n) \rightarrow L$ has *telescoping order* k if $|D_n| = \Theta(n^{-(k+1)})$.

Remark 2.2. The phrase “order k ” here refers to the decay rate $D_n \asymp n^{-(k+1)}$. By Lemma 1.2, this corresponds to an $O(N^{-k})$ rate for the approximation error $|x_N - L|$ (with a signed asymptotic, as discussed in Remark 1.3).

2.2 Asymptotic expansion of $\log x_n$

Write $u_n := f(1/n) - 1 = \alpha_1/n + \alpha_2/n^2 + \cdots$. Then $\log(1 + u_n) = u_n - \frac{u_n^2}{2} + \frac{u_n^3}{3} - \cdots$ and $\log x_n = g(n) \log(1 + u_n)$. Expanding in powers of $1/n$,

$$\log x_n = c_0 + \frac{A_1}{n} + \frac{A_2}{n^2} + \frac{A_3}{n^3} + \frac{A_4}{n^4} + O(n^{-5}), \quad (3)$$

where $c_0, A_1, A_2, \dots$ are explicit polynomials in $\{\alpha_j, \beta, \gamma\}$ (given in §4).

2.3 From $\log x_n$ to D_n

The following lemma is the bridge from the logarithmic expansion to the difference asymptotic. Note the additional regularity hypothesis: a bare $O(n^{-(k+2)})$ remainder cannot be differenced as though it were smooth (the remainder’s finite difference need not be $O(n^{-(k+3)})$).

Lemma 2.3 (Log-expansion implies difference asymptotic). *Suppose that for some integer $k \geq 0$ the sequence admits an asymptotic expansion in powers of $1/n$,*

$$\log x_n = c + \frac{A}{n^{k+1}} + \frac{B}{n^{k+2}} + O(n^{-(k+3)}), \quad A \neq 0. \quad (4)$$

Then

$$x_n = e^c \left(1 + \frac{A}{n^{k+1}} + O(n^{-(k+2)}) \right), \quad D_n := x_{n+1} - x_n = -(k+1)e^c A n^{-(k+2)} + O(n^{-(k+3)}).$$

Proof. Exponentiating (4) and using $\exp(z) = 1 + z + O(z^2)$ gives $x_n = e^c (1 + A n^{-(k+1)} + B n^{-(k+2)} + O(n^{-(k+3)}))$ for $k \geq 1$, since $(n^{-(k+1)})^2 = O(n^{-(k+3)})$; for $k = 0$ the n^{-2} coefficient instead becomes $B + A^2/2$, which is immaterial at the order computed. Writing $x_n = e^c (1 + A n^{-(k+1)} + R_n)$ we thus have $R_n = B n^{-(k+2)} + O(n^{-(k+3)})$, so

$$\Delta R_n = B[(n+1)^{-(k+2)} - n^{-(k+2)}] + O(n^{-(k+3)}) = O(n^{-(k+3)})$$

(the mean value theorem gives the first term, and the triangle inequality bounds the remainder). Hence

$$D_n = e^c A [(n+1)^{-(k+1)} - n^{-(k+1)}] + O(n^{-(k+3)}),$$

and $(n+1)^{-(k+1)} - n^{-(k+1)} = -(k+1)n^{-(k+2)} + O(n^{-(k+3)})$ by the mean value theorem, which completes the proof. $\square$

Design principle. To achieve telescoping order k in the sense of Definition 2.1, enforce $c_0 = c$ and $A_1 = \dots = A_{k-1} = 0$, leaving $A_k \neq 0$. (Indeed, if the first nonzero coefficient is A_k , then $\log x_n = c + A_k n^{-k} + O(n^{-(k+1)})$, and Lemma 2.3 with the relabeled index $k-1$ gives $D_n = -k e^c A_k n^{-(k+1)} + O(n^{-(k+2)}) = \Theta(n^{-(k+1)})$, i.e. order k .)

3 Order-1: Classical Cancellation

Proposition 3.1 (Two-parameter family). *For $x_n = (1 + a/n)^{bn}$ with real a, b and n sufficiently large that the base is positive,*

$$\lim_{n \rightarrow \infty} x_n = e^{ab}, \quad D_n = \frac{e^{ab} a^2 b}{2n^2} + O(n^{-3}).$$

Consequently $D_n = O(n^{-3})$ if and only if $a = 0$ or $b = 0$ (in which case $x_n \equiv 1$); for $a, b \neq 0$ the difference order is exactly $\Theta(n^{-2})$.

Proof. Since $\log(1 + a/n) = a/n - a^2/(2n^2) + O(n^{-3})$,

$$\log x_n = bn \log \left(1 + \frac{a}{n} \right) = ab - \frac{a^2 b}{2n} + \frac{a^3 b}{3n^2} + O(n^{-3}).$$

Hence $x_n = e^{ab} (1 - \frac{a^2 b}{2n} + O(n^{-2}))$, and Lemma 2.3 with $k = 0$ and $A = -a^2 b/2$ gives $D_n = e^{ab} \frac{a^2 b}{2n^2} + O(n^{-3})$. $\square$

Remark 3.2 (Correction of an earlier draft). An earlier version of this paper stated $D_n = e^{ab} ab(a-b)/(2n^2) + O(n^{-3})$ and concluded that the leading term cancels at $a = b$. That is incorrect: the leading coefficient is $a^2 b/2$, not $ab(a-b)/2$, and it does not vanish at $a = b$. Numerically, for Euler's sequence ($a = b = 1$) one has $n^2 D_n / e \rightarrow 1/2$, i.e. $D_n = \Theta(n^{-2})$, not $O(n^{-3})$. The corrected statement is Proposition 3.1; see also Table 3.

4 Explicit Order-3 Constructions

4.1 Quadratic base, affine exponent

Consider

$$x_n = \left(1 + \frac{\alpha_1}{n} + \frac{\alpha_2}{n^2}\right)^{\beta n + \gamma}.$$

Let $u_n = \alpha_1/n + \alpha_2/n^2$. Expanding $\log(1 + u_n)$ and multiplying by $\beta n + \gamma$ yields

$$c_0 = \alpha_1 \beta, \tag{5}$$

$$A_1 = \alpha_1 \gamma + \beta \alpha_2 - \frac{\alpha_1^2 \beta}{2}, \tag{6}$$

$$A_2 = \gamma \alpha_2 - \frac{\alpha_1^2 \gamma}{2} + \frac{\alpha_1^3 \beta}{3} - \alpha_1 \alpha_2 \beta, \tag{7}$$

$$A_3 = \gamma \left(\frac{\alpha_1^3}{3} - \alpha_1 \alpha_2 \right) + \beta \left(\alpha_1^2 \alpha_2 - \frac{\alpha_1^4}{4} - \frac{\alpha_2^2}{2} \right), \tag{8}$$

$$A_4 = \gamma \left(\alpha_1^2 \alpha_2 - \frac{\alpha_1^4}{4} - \frac{\alpha_2^2}{2} \right) + \beta \left(\frac{\alpha_1^5}{5} - \alpha_1^3 \alpha_2 + \alpha_1 \alpha_2^2 \right). \tag{9}$$

(The A_4/n^4 term is quoted for completeness; it controls the subleading correction and is used in §4.3.)

Remark 4.1 (Correction of an earlier draft). An earlier version printed the β -part of (8) as $\beta(\alpha_1^4/4 - \alpha_1^2 \alpha_2/2)$, with two incorrect signs and the $-\alpha_2^2/2$ term missing. The correct form is (8). As a check, substituting the branch parameters into the incorrect form gives $c(9 - 4\sqrt{3})/18$, which agrees neither with the correct value nor with the earlier draft's own stated constant.

4.2 Constraint system and closed-form solutions

To achieve telescoping order 3 we require $c_0 = c$, $A_1 = 0$, $A_2 = 0$. We have four parameters $\{\alpha_1, \alpha_2, \beta, \gamma\}$ and three constraints. Fix $\alpha_1 = 1$ and $\beta = c$ to satisfy $c_0 = c$. The remaining constraints yield

$$\gamma + c\alpha_2 - \frac{c}{2} = 0, \quad \gamma\alpha_2 - \frac{\gamma}{2} + \frac{c}{3} - c\alpha_2 = 0.$$

From the first, $\gamma = c(\frac{1}{2} - \alpha_2)$. Substituting into the second gives $\alpha_2^2 = \frac{1}{12}$, so $\alpha_2 = \pm \frac{\sqrt{3}}{6}$. Thus we obtain two explicit branches:

$$\text{Family I: } \alpha_2 = +\frac{\sqrt{3}}{6}, \quad \gamma = c\left(\frac{1}{2} - \frac{\sqrt{3}}{6}\right); \quad \text{Family II: } \alpha_2 = -\frac{\sqrt{3}}{6}, \quad \gamma = c\left(\frac{1}{2} + \frac{\sqrt{3}}{6}\right).$$

4.3 Explicit formula for A_3 and nonvanishing

Substituting $\alpha_1 = 1$, $\beta = c$ and the corresponding γ values into (8) gives

$$A_3^{(\text{I})} = \frac{c(2\sqrt{3} - 3)}{72}, \quad A_3^{(\text{II})} = -\frac{c(3 + 2\sqrt{3})}{72}.$$

Since $2\sqrt{3} - 3 > 0$ and $3 + 2\sqrt{3} > 0$, we have $A_3^{(\cdot)} \neq 0$ whenever $c \neq 0$ (with opposite signs on the two branches). By Lemma 2.3 (with $k = 2$ and $A = A_3$),

$$D_n \sim -\frac{3e^c A_3}{n^4}.$$

For completeness, substituting into (9) gives the subleading coefficients

$$A_4^{(\text{I})} = \frac{c(39 - 25\sqrt{3})}{720}, \quad A_4^{(\text{II})} = \frac{c(39 + 25\sqrt{3})}{720},$$

which quantify the $O(n^{-5})$ corrections to D_n and address Open Problem 1 below.

Remark 4.2 (Correction of an earlier draft). An earlier version stated $A_3^{(\text{I})} = c(9 - 5\sqrt{3})/36$ and $A_3^{(\text{II})} = c(9 + 5\sqrt{3})/36$, together with the multiplier $-4e^c$ in (2). Both are incorrect: the constants are as above, and the multiplier is $-3e^c$ (as required by Lemma 2.3 with $k = 2$). The paper’s own numerical tables already reflected the corrected values; only the closed forms were in error.

4.4 Proof of the main theorem

Proof of Theorem 1.5. With the parameters above, we have $c_0 = c$, $A_1 = 0$, $A_2 = 0$ and the stated $A_3 \neq 0$. Hence

$$\log x_n = c + \frac{A_3}{n^3} + \frac{A_4}{n^4} + O(n^{-5}),$$

which is of the form (4) with $k = 2$, $A = A_3$, $B = A_4$. Lemma 2.3 yields $D_n \sim -3e^c A_3 n^{-4}$, so $|D_n| = \Theta(n^{-4})$. Finally, Lemma 1.2 implies the corresponding error rate $|x_N - e^c| = \Theta(N^{-3})$. $\square$

5 Numerical Validation

We implement $x_n^{(\cdot)}$ at 150-digit precision using `mpmath` and compute $D_n = x_{n+1} - x_n$ for $100 \leq n \leq 10000$. Log–log regression over the grid $n = 100, 125, \dots, 10000$ yields

$$\text{Family I: } \log |D_n| = -3.9973 \log n - 2.9696, \quad R^2 = 0.9999997,$$

$$\text{Family II: } \log |D_n| = -3.9969 \log n - 0.3391, \quad R^2 = 0.9999996,$$

consistent with the predicted n^{-4} decay. (The fitted slopes differ from -4 by a small finite-range curvature; the intercepts likewise carry an $O(n^{-1})$ correction. The numbers above are regenerated by the script in Appendix A.)

5.1 A log–log “figure” (self-contained, text-only)

A literal log–log plot of $|D_n|$ versus n would appear as an approximately straight line with slope near -4 for both branches. The regression above provides a compact self-contained surrogate for the visual.

Detailed convergence for $c = 1$ (target $e \approx 2.71828$):

n	$ x_n^{(\text{I})} - e $	$ D_n^{(\text{I})} $	$ x_n^{(\text{II})} - e $	$ D_n^{(\text{II})} $
100	1.74×10^{-8}	5.09×10^{-10}	2.41×10^{-7}	7.06×10^{-9}
500	1.40×10^{-10}	8.36×10^{-13}	1.95×10^{-9}	1.16×10^{-11}
1000	1.75×10^{-11}	5.24×10^{-14}	2.44×10^{-10}	7.29×10^{-13}
5000	1.40×10^{-13}	8.40×10^{-17}	1.95×10^{-12}	1.17×10^{-15}

Table 1: Convergence of $x_n^{(\cdot)}$ to e ($c = 1$) at 150-digit precision.

5.2 Generality check: additional values of c

Table 2 reports representative errors and differences for $c = 2$ and $c = -1$.

c	n	$ x_n^{(\text{I})} - e^c $	$ D_n^{(\text{I})} $	$ D_n^{(\text{II})} $
2	1000	9.52×10^{-11}	2.85×10^{-13}	3.97×10^{-12}
2	5000	7.62×10^{-13}	4.57×10^{-16}	6.36×10^{-15}
-1	1000	2.37×10^{-12}	7.09×10^{-15}	9.87×10^{-14}
-1	5000	1.90×10^{-14}	1.14×10^{-17}	1.58×10^{-16}

Table 2: Illustrative generality check for $c = 2$ and $c = -1$ at high precision.

6 Comparison with Existing Methods

6.1 Convergence-rate comparison

By Lemma 1.2, if D_n has the signed asymptotic $D_n = Kn^{-(k+1)} + O(n^{-(k+2)})$ with $K \neq 0$ (as holds for every family compared below), then $|x_N - L| = \Theta(N^{-k})$. Thus, for a target error ε , one expects $N \asymp \varepsilon^{-1/k}$ in such telescoping constructions.

Method	Difference decay D_n	Error decay $ x_N - L $	N for $ x_N - L < \varepsilon$
Euler $(1 + 1/n)^n$	$\Theta(n^{-2})$	$\Theta(N^{-1})$	$\asymp \varepsilon^{-1}$
Classical $(1 + c/n)^{cn}$	$\Theta(n^{-2})$	$\Theta(N^{-1})$	$\asymp \varepsilon^{-1}$
Order-3 (this work)	$\Theta(n^{-4})$	$\Theta(N^{-3})$	$\asymp \varepsilon^{-1/3}$

Table 3: Telescoping decay of differences and implied error rates (parameterized rows: fixed $c \neq 0$).

Remark 6.1. An earlier draft listed the middle row as $\Theta(n^{-3})/\Theta(N^{-2})$ with target e^c ; that is incorrect. For $(1 + c/n)^{cn}$ the limit is e^{c^2} and the difference order is $\Theta(n^{-2})$ (Proposition 3.1); at $c = 1$ it coincides with Euler’s sequence in the first row. To obtain $O(N^{-2})$ error decay one must instead *enlarge* the family (e.g. by an affine exponent); by Proposition 3.1 no nonconstant member of $(1 + a/n)^{bn}$ has order 2, so this is not pursued here.

6.2 Computational cost and limitations (pedagogical framing)

We make no competitive complexity claim. For a fixed working precision the construction attains tail error $< \varepsilon$ at index $N \asymp \varepsilon^{-1/3}$, so a single evaluation of x_N suffices; the telescoping identity means one need not sum all N differences. We deliberately state no bit-complexity bound: the cost of a real power a^b at precision p depends on the underlying exp / log algorithm and is not $O(\log n)$ arithmetic operations in general. This is not intended as a competitive high-precision algorithm for computing e^c (where AGM-based and related methods converge far faster). Instead, the value of the construction is primarily structural and pedagogical: it provides a compact closed-form example where higher-order cancellation can be derived, proved, and numerically verified end-to-end using only elementary expansions and a small parameter system. In particular, it is well-suited for illustrating (i) how asymptotic expansions control discrete differences, (ii) how cancellation constraints translate into algebraic equations, and (iii) how regression on log-log data can detect decay exponents in practice.

7 Open Problems

1. **Subleading corrections:** the $O(n^{-5})$ and higher terms are now available in closed form through A_4 (§4.3); deriving the complete expansion would explain the effective constants observed at moderate n .
2. **Order-4 construction:** find explicit closed-form solutions yielding $D_n = \Theta(n^{-5})$, likely requiring at least a cubic base or a richer exponent structure. A natural next ansatz is $x_n = (1 + a_1/n + a_2/n^2 + a_3/n^3)^{cn+\gamma}$, which introduces additional degrees of freedom.
3. **Other constants (including γ):** extend the framework to other constants using appropriate base families. A concrete target is the Euler–Mascheroni constant $\gamma = \lim_{n \rightarrow \infty} (H_n - \log n)$, $H_n = \sum_{k \leq n} 1/k$; one may try to match coefficients in $y_n \approx \gamma + \frac{1}{2n} - \frac{1}{12n^2} + \frac{1}{120n^4} + \dots$ to enforce cancellations in Δy_n ; for accurate asymptotic formulas for the special functions underlying such constant constructions, see [10, 11].
4. **Non-power forms:** investigate telescoping sequences outside the power-form ansatz.
5. **Optimality:** characterize the maximal telescoping order achievable under bounded algebraic complexity.

8 Conclusion

We presented an explicit closed-form two-branch construction of sequences converging to e^c with telescoping order 3 (i.e. $|D_n| = \Theta(n^{-4})$), obtained by enforcing coefficient-cancellation constraints within a quadratic-base, affine-exponent family, together with explicit signed leading constants $A_3^{(\cdot)}$ and the implied $\Theta(N^{-3})$ error rate. The broader point is that explicit telescoping families provide clean examples where difference-asymptotic control translates directly into error rates, making them useful both for analysis and for didactic illustration of convergence acceleration ideas [6].

Note on this revision

This is revision V5. Relative to V4 it corrects: the A_3 coefficient formula (8); the constants $A_3^{(\cdot)}$ and the multiplier $3e^c$ in (2); the classical baseline (Proposition 3.1, Remark 3.2); the design-principle indexing; the hypothesis of Lemma 2.3 and the signed-asymptotic caveat (Remark 1.3); Table 3’s middle row; the $c = 0$ exception in the abstract; and several bibliographic entries. All numerical tables were regenerated from the committed driver (Appendix A); the one mis-rounded entry in the earlier Table 1 ($8.41 \times 10^{-17} \rightarrow 8.40 \times 10^{-17}$) is corrected.

Acknowledgments

The author acknowledges the use of AI-assisted tools as editorial and programming support during the preparation of this manuscript. All mathematical derivations, proofs, asymptotic expansions, and numerical computations reported in the paper were verified by the author against an independent verification driver shipped with the paper.

Declaration of Competing Interest

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

The author reports no funding for this research.

References

- [1] R. P. Brent, *Fast multiple-precision evaluation of elementary functions*, J. ACM **23**(2) (1976), 242–251.
- [2] J. W. L. Glaisher, *On the history of Euler’s constant*, Messenger of Mathematics **1** (1872), 25–30.
- [3] J. C. Adams, *Note on the value of Euler’s constant; likewise on the values of the Napierian logarithms of 2, 3, 5, 7, and 10*, Proc. Royal Soc. London **27** (1878), 88–94.
- [4] U. G. Mitchell and M. Strain, *The number e* , Osiris **1** (1936), 476–496.
- [5] R. Sedgewick and P. Flajolet, *An Introduction to the Analysis of Algorithms*, booksite Chapter 4, *Asymptotic Approximations*, <https://aofa.cs.princeton.edu/40asymptotic/>.
- [6] NIST Digital Library of Mathematical Functions, §3.9 *Acceleration of Convergence*, <https://dlmf.nist.gov/3.9>.
- [7] R. M. Corless, *Devilish tricks for sequence acceleration*, Maple Transactions **3**(1) (2023), DOI <https://doi.org/10.5206/mt.v3i1.14777>.
- [8] C. J. Oates, T. Karvonen, A. L. Teckentrup, M. Strocchi, and S. A. Niederer, *Probabilistic Richardson extrapolation*, J. Roy. Statist. Soc. Ser. B **87**(2) (2025), 457–479, DOI <https://doi.org/10.1093/jrssb/qkae098> (published online 26 December 2024).
- [9] C. J. Oates, T. Karvonen, A. L. Teckentrup, M. Strocchi, and S. A. Niederer, *Probabilistic Richardson extrapolation*, arXiv:2401.07562 (2024), <https://arxiv.org/abs/2401.07562>.
- [10] Z.-H. Yang and J.-F. Tian, *An accurate approximation formula for gamma function*, J. Inequalities & Applications **2018**, Article 56, DOI <https://doi.org/10.1186/s13660-018-1646-6>.
- [11] Z.-H. Yang and J.-F. Tian, *An accurate approximation formula for gamma function*, arXiv:1712.08051 (2017), <https://arxiv.org/abs/1712.08051>.
- [12] J. Bald, *Telescoping representations of classical iterative methods and constants*, Zenodo (2025), version DOI <https://doi.org/10.5281/zenodo.17605159>; concept DOI <https://doi.org/10.5281/zenodo.17605158>.

A Reproducibility Code (Python)

This appendix provides a self-contained script reproducing the high-precision numerical validation in §5: computation of $x_n^{(I)}$, $x_n^{(II)}$, differences $D_n = x_{n+1} - x_n$, log-log slope estimates of $|D_n|$ versus n , and sample convergence values for $c = 1, 2, -1$.

Dependencies. `mpmath` (arbitrary precision). The full verification driver (`code/verify_hoc.py`), which additionally checks the symbolic coefficients (8)–(9) and *asserts every claimed sign and constant*, is available in the reproduction package.

```
import mpmath as mp

def xI(n, c):
    n = mp.mpf(n); s3 = mp.sqrt(3)
    return mp.power(1 + 1/n + s3/(6*n**2), c*n + c*(mp.mpf(1)/2 - s3/6))

def xII(n, c):
    n = mp.mpf(n); s3 = mp.sqrt(3)
    return mp.power(1 + 1/n - s3/(6*n**2), c*n + c*(mp.mpf(1)/2 + s3/6))

def slope(f, c, n0=100, n1=10000, step=25):
    ns = list(range(n0, n1+1, step))
    xs = [mp.log(n) for n in ns]
    ys = [mp.log(mp.fabs(f(n+1, c) - f(n, c))) for n in ns]
    m = len(ns); mx = mp.fsum(xs)/m; my = mp.fsum(ys)/m
    cov = mp.fsum([(xs[i]-mx)*(ys[i]-my) for i in range(m)])
    var = mp.fsum([(xs[i]-mx)**2 for i in range(m)])
    return cov/var, my - (cov/var)*mx

if __name__ == "__main__":
    mp.mp.dps = 150
    c = mp.mpf(1); target = mp.e**c
    print("slope(I) :", slope(xI, c))
    print("slope(II):", slope(xII, c))
    for n in [100, 500, 1000, 5000]:
        print(n, mp.fabs(xI(n,c)-target), mp.fabs(xI(n+1,c)-xI(n,c)),
              mp.fabs(xII(n,c)-target), mp.fabs(xII(n+1,c)-xII(n,c)))
    # leading-constant check: n^4 * D_n -> -3 e^c A3
    for f, A3 in [(xI, (2*mp.sqrt(3)-3)/72), (xII, -(3+2*mp.sqrt(3))/72)]:
        n = 10**5
        print("n^4*D_n ->", (f(n+1,c)-f(n,c))*n**4, " predicted", -3*mp.e*A3)
```

B Verification Summary

The companion driver (`code/verify_hoc.py`) regenerates every numeral in this paper and asserts each claimed sign and constant (and exits with nonzero status on any failed assertion); it reports **ALL ASSERTIONS PASS**. Symbolically it confirms (5)–(9) (including the general A_4 identity), $A_1 = A_2 = 0$ on both branches, and the constants $A_3^{(I)} = c(2\sqrt{3}-3)/72$, $A_3^{(II)} = -c(3+2\sqrt{3})/72$, $A_4^{(I,II)} = c(39 \mp 25\sqrt{3})/720$. Numerically it confirms $\lim x_n = e^c$, $n^4 D_n \rightarrow -3e^c A_3$ and $n^3(x_n - e^c) \rightarrow e^c A_3$ with the correct signs, every tabulated value in Tables 1–2, the regression slopes and R^2 , and the corrected classical baseline $D_n \sim e^{ab} a^2 b / (2n^2)$. The exact algebraic core (the branch cancellations and the constants A_3, A_4) is machine-checked in Lean 4 (`lean/ExpHOC`); the gate reports **PASS** (13 theorems, standard axioms only).

The complete reproduction package — manuscript source, driver and its saved output, the

Lean development, and the independent audit reports — is available at <https://github.com/jpbald93/telescoping-exponential-approximants>.